

# Global Art Market Analytics

A simple look at artwork valuations, market segments and estimate performance

# What I wanted to understand

- What factors are associated with higher recorded artwork prices?
- How do prices differ across countries, genres and artist tiers?
- How closely do recorded prices align with pre-sale estimates?
- How do artwork prices change over time?

## **Dataset:** Contemporary Art Auction Dataset

**Source:** Lee et al. (2024), *Scientific Reports*

**Authors:** Kangsan Lee, Jaehyuk Park, Sam Goree, David J. Crandall, Yong-Yeol Ahn

# Market overview

**34,200**

Artwork records

**\$17,691**

Median recorded price

**43.1%**

Above estimate

- Recorded prices vary considerably across artist tiers, genres and market segments.
- Median price is used as the main price measure because a small number of very high-priced works can pull the mean upward.
- 34,200 artwork records were analyzed.

# GLOBAL ART MARKET ANALYTICS

Country

All

year

1996

2012

Artwork Count

34K

Median Price

\$17.69K



Total Recorded Value

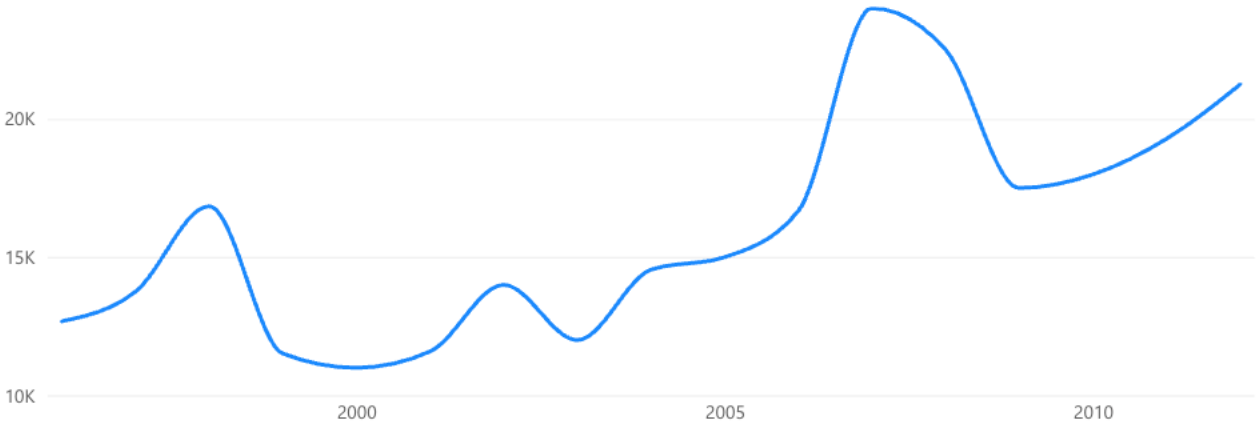
£2.496799388bn

Above Estimate

43.13 %

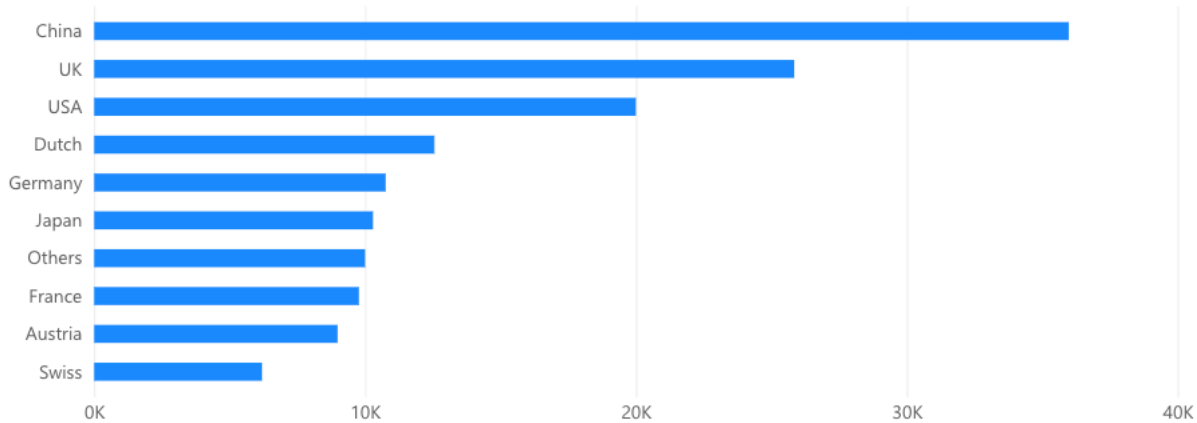
## Median Recorded Price by Year

Median Price by year



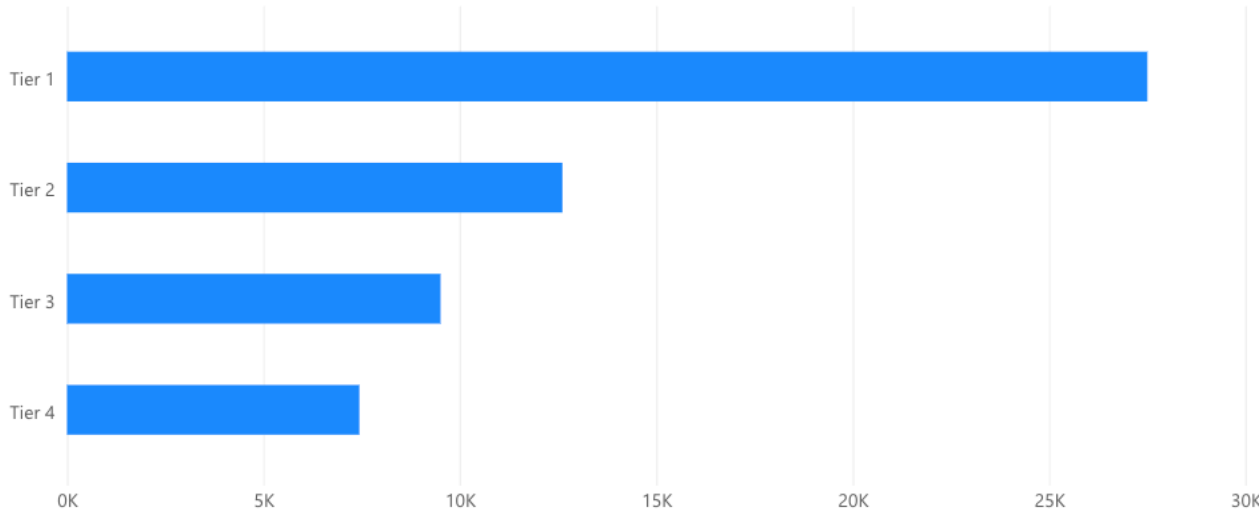
## Median Recorded Price by Country

Median Price by Country



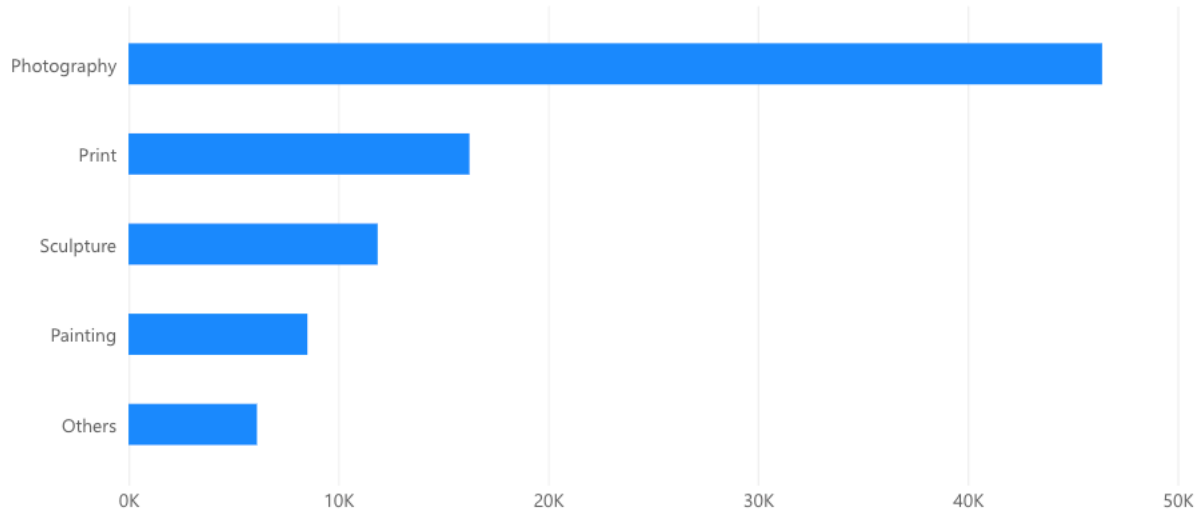
## Median Recorded Price by Tier

Median Price by Tier



## Median Recorded Price by Genre

Median Price by Genre



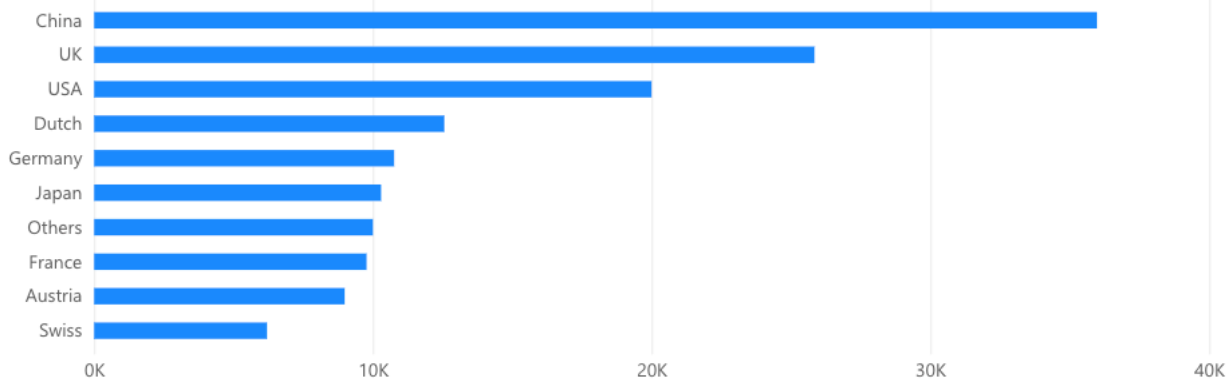
# What stands out across market segments

- Artist tier is strongly associated with typical recorded price. Tier 1 has a median of \$27,500 versus \$7,443 for Tier 4.
- Pricing also differs by genre. Photography has the highest median recorded price in this dataset at \$46,398.
- Country-level prices vary as well, although sample sizes differ substantially between countries.
- These patterns suggest that market positioning and segment characteristics matter when benchmarking artwork valuations.

# MARKET SEGMENTS

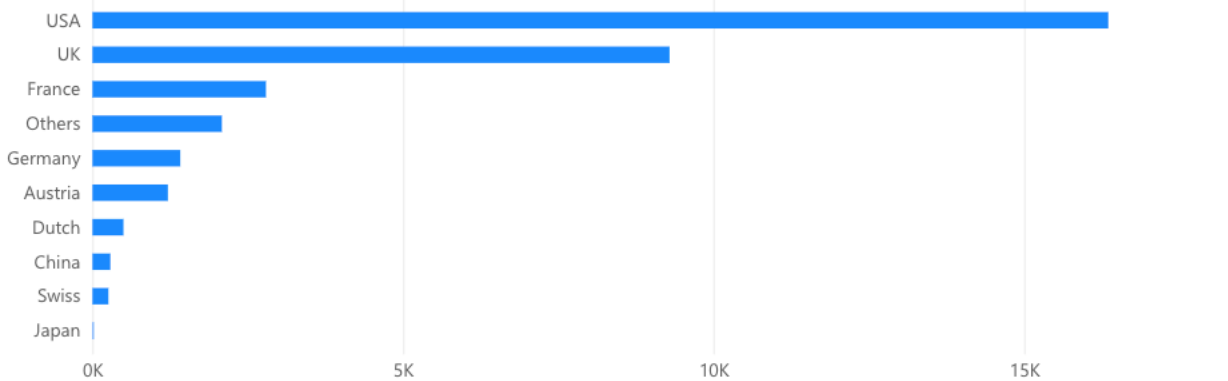
Median Recorded Price by Country

Median Price by Country



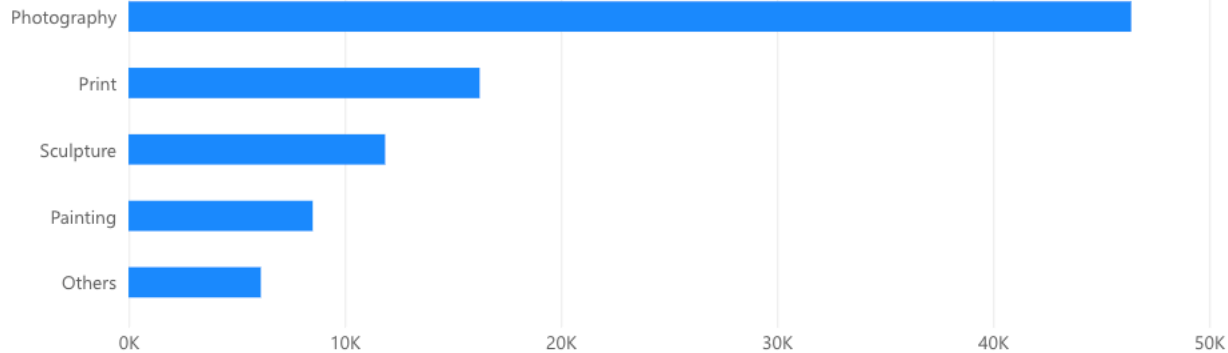
Artwork Records by Country

Artwork Count by Country



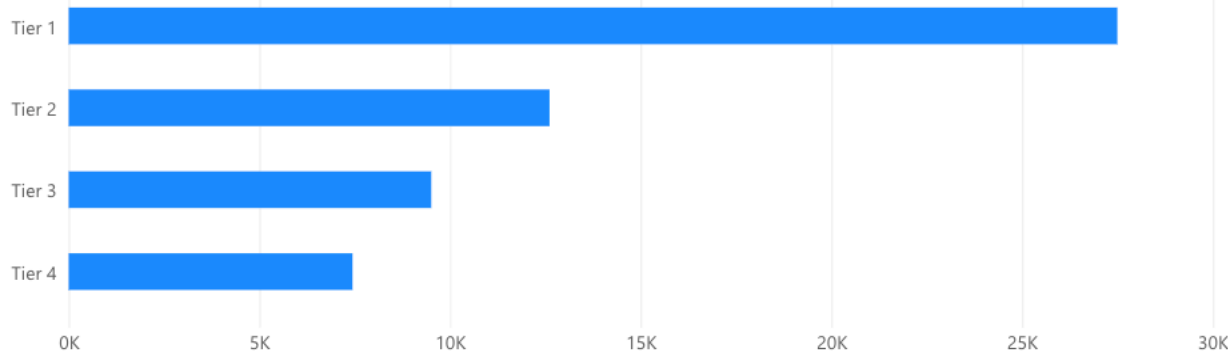
Median Recorded Price by Genre

Median Price by Genre



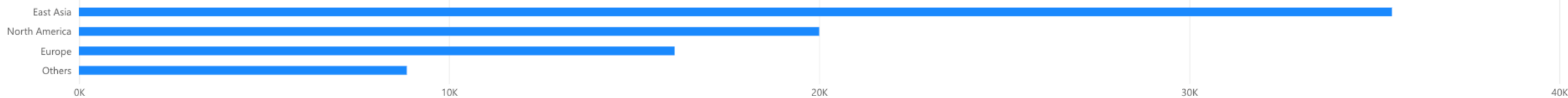
Median Recorded Price by Artist Tier

Median Price by Tier



Median Recorded Price by Continent

Median Price by Continent



# Estimate performance

- 43.1% of recorded prices were above the stated estimate range.
- 41.0% were within the estimate range and 15.9% were below it.
- The median recorded price was about 1.09× the estimate midpoint.
- Pre-sale estimates are useful benchmarks, but recorded prices can move meaningfully away from expectations.

# ESTIMATE PERFORMANCE

Median Estimate Ratio

1.09

Above Estimate

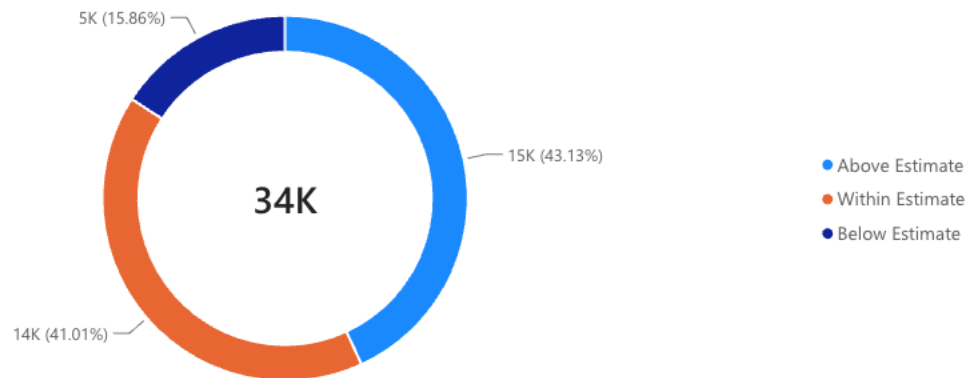
43.13 %

Artwork Count

34K

## Recorded Price vs Estimate Range

Artwork Count by estimate\_status



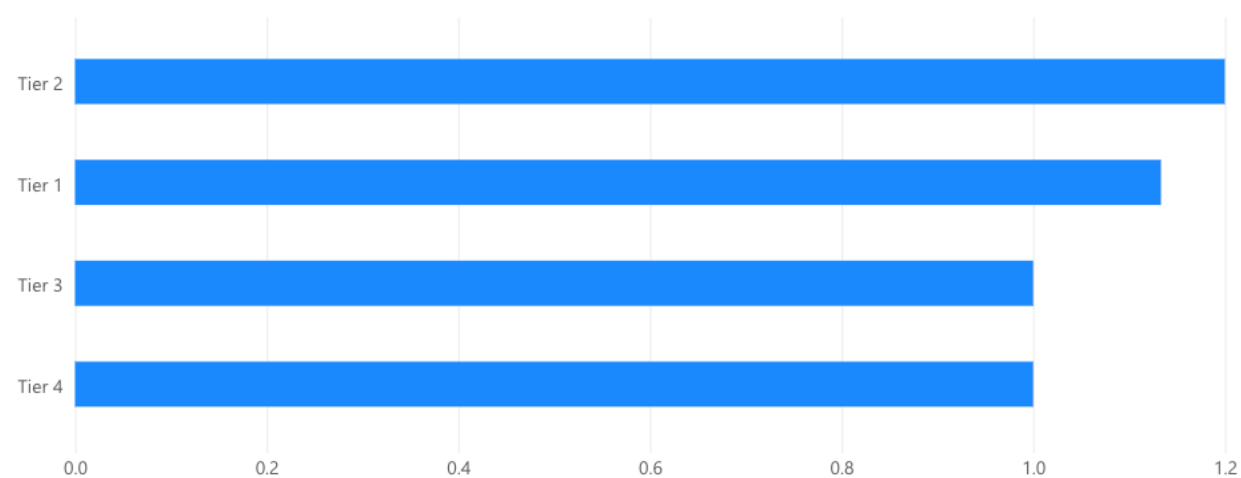
## Median Estimate Ratio

by year



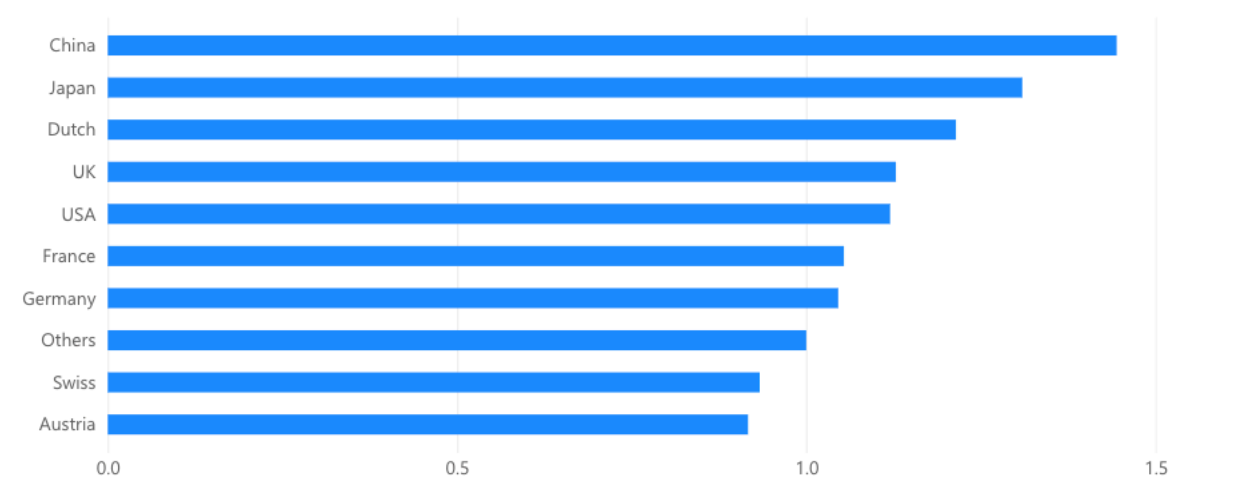
## Median Estimate Ratio

by Tier



## Median Estimate Ratio

by Country



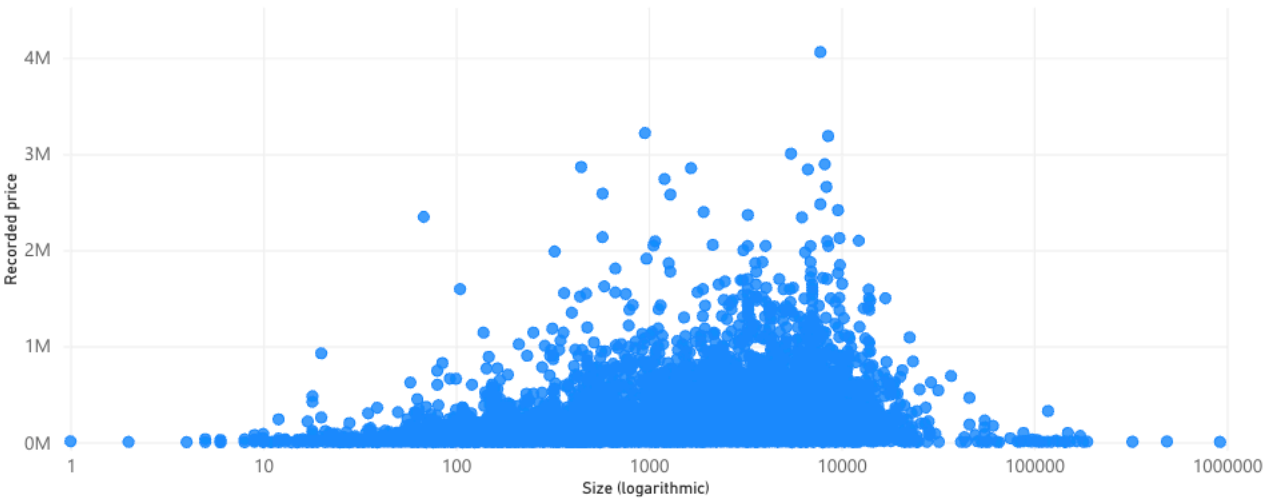
# Artwork characteristics

- Artwork size has a moderate positive relationship with recorded price (Spearman correlation  $\approx 0.41$ ).
- The relationship is not strong enough to treat size as a standalone explanation of price.
- Genre and artist tier continue to show clear differences when looking at typical prices.
- For valuation analysis, physical characteristics are therefore better treated as one part of a broader set of factors.

# ARTWORK CHARACTERISTICS

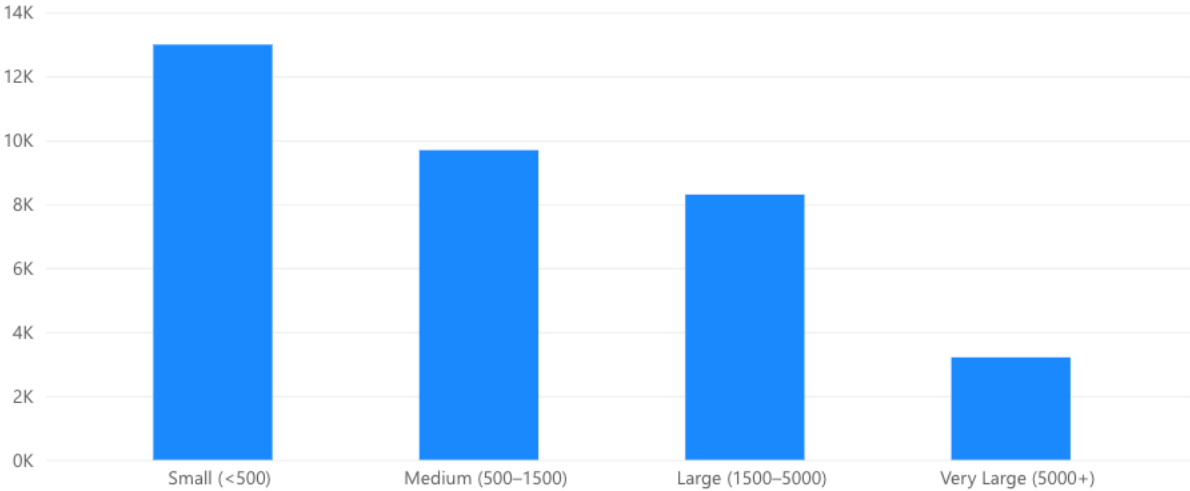
Artwork Size vs Recorded Price

Sum of size\_inchsqr and Recorded price by case\_id



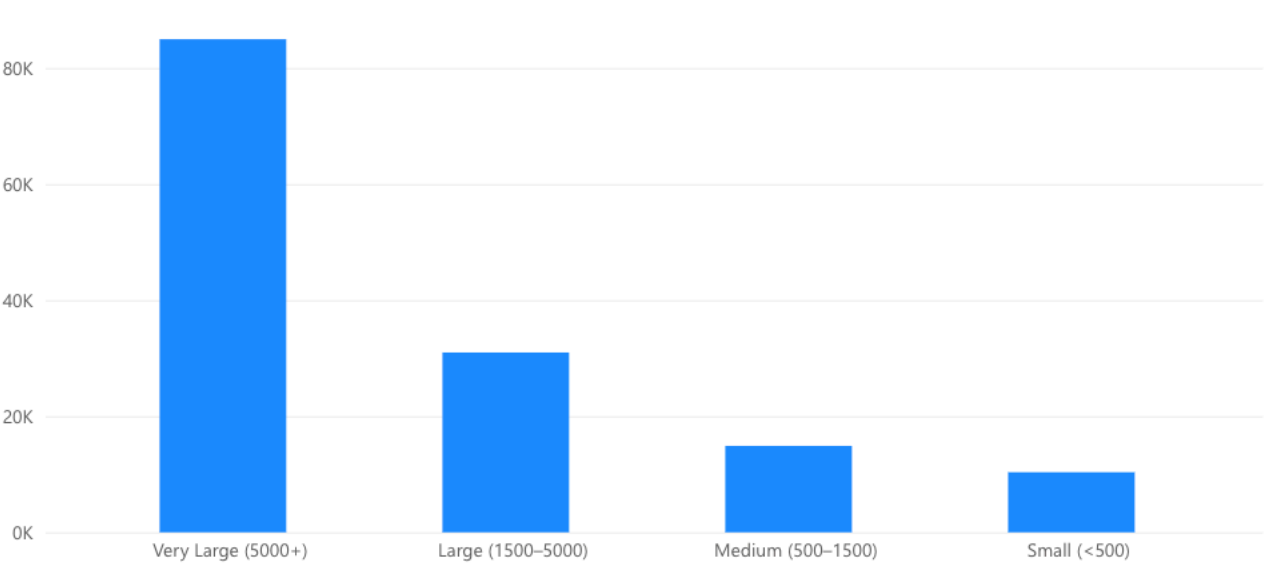
Artwork Count

by Size Bucket



Median Price

by Size Bucket



Median Recorded Price by Genre and Artist Tier

| Genre       | Tier 1    | Tier 2    | Tier 3    | Tier 4    | Total     |
|-------------|-----------|-----------|-----------|-----------|-----------|
| Others      | 8,125.00  | 7,507.50  | 3,362.00  | 5,044.00  | 6,124.00  |
| Painting    | 9,808.00  | 8,464.50  | 7,964.00  | 6,720.50  | 8,525.00  |
| Photography | 82,805.00 | 25,868.50 | 16,047.00 | 14,303.50 | 46,398.00 |
| Print       | 20,000.00 | 9,706.00  | 8,327.00  | 7,209.00  | 16,250.00 |
| Sculpture   | 24,000.00 | 10,860.00 | 5,287.50  | 5,367.00  | 11,875.00 |
| Total       | 27,500.00 | 12,610.00 | 9,508.50  | 7,443.00  | 17,691.00 |

# Pricing changes over time

- The dataset shows a noticeable peak around 2007, when median recorded price reached \$24,000.
- Prices changed in the years that followed, showing that historical price levels are not static.
- The broader art market was also unusually strong around this period, followed by a correction during the 2008 financial crisis.
- Time-aware benchmarks are therefore more useful than relying on a single historical average.

# Assumptions tested through Data

Established artists command higher prices

**Supported** — Tier 1 median was \$27.5K vs \$7.44K for Tier 4

Bigger artworks are always more valuable

**Challenged** — size had only a moderate correlation with price

Higher-priced tiers should always beat estimates

**Challenged** — Tier 2 had higher estimate-relative performance than Tier 1

Art prices steadily increase over time

**Challenged** — prices fluctuated substantially, including a 2007 peak and subsequent decline

# Key takeaways

- Artist tier and genre show substantial segmentation in typical recorded prices.
- Country and continent provide useful market context, but comparisons should account for differences in sample size.
- Art pieces are highly skewed.
- Estimate-to-recorded-price variance provides a simple way to monitor how market outcomes differ from expectations.
- Artwork size is informative, but valuation is clearly a multi-factor problem.
- Overall, the dashboard can be used as a starting point for segment benchmarking and deeper valuation analysis.

# Tools and approach

- Python: data cleaning, category conversion, descriptive statistics and correlation analysis.
- Power BI: KPI measures, visual analysis and dashboard design.
- Main measures: artwork count, median recorded price, total recorded value, above-estimate percentage and median estimate ratio.
- The analysis is descriptive: the results show associations and patterns rather than proving causation.